

Chapter II

Brief Solutions

1 INPhO-2006

1. (a) $a=0$
 (b) $f_s = 5 \text{ N.}$ and $T \cong 10 \text{ N.}$
2. (a) $a = \frac{\mu g}{3}$
 (b) $t^* = \frac{3}{7} \cdot \frac{R \omega_0}{\mu g}$
 (c) $t_{tot} = \frac{R \omega_0}{\mu g}$
3. (a) $\eta = 1 - \frac{T_c - T_d}{T_b - T_a}$
 (b) $\eta = 1 - \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$
 (c)

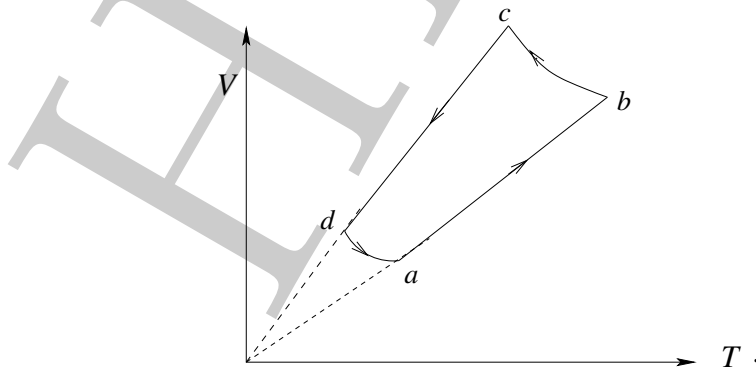


Figure 1: V - T diagram

- (d) $\eta_c = 1 - \frac{T_d}{T_b}$,
 Carnot cycle has higher efficiency than the given cycle.

4. Amplitude of the image = 2.78 cm
 The phase difference = 0
 Angle made by the virtual image with the principle axis = 30.26°
5. (a) $\mu' = \frac{\pi\omega\sigma R^4}{4}$ and along the positive x -direction.
 (b) $B = \frac{\mu_0\omega\sigma}{2} \left[\frac{R^2 + 2a^2}{\sqrt{R^2 + a^2}} - 2a \right]$ and along the positive x -direction.
 (c) $B = \frac{\mu_0\omega\sigma R^4}{8a^3}$ and along the positive x -direction.
 (d) $F = -\frac{3\mu\mu_0\omega\sigma R^4}{8a^4}$ and direction dictated by the direction of $\vec{\mu}$.
6. (a) $F = 4\pi np \left[\frac{\delta^2}{2} - \frac{\delta^4}{4R^2} \right]$ and direction upwards.
 (b) $m = 6.80 \times 10^{-7}$ kg.
7. (a) $\lambda = 2080 \text{ \AA}$. This wavelength belongs to ultraviolet region of the electromagnetic spectrum.
 (b) Maximum Kinetic Energy $K_{max} = 2.75 \text{ eV}$
 De Broglie wavelength $\lambda_{dB} = 7.4 \text{ \AA}$
8. (a) $y \geq 10^{-18}$
 (b) $F = \frac{\eta e^2 y^2 r}{3\epsilon_0 m_p}$
 (c) $H = 1.8 \times 10^{-17} \text{ s}^{-1}$
 (d) $V = V_0 e^{3Ht}$
 (e) The numerical value of H does not agree with the known value of Hubble's constant. This is one of the reasons we may discard the Lyttleton Bondi model.